NewsBot Intelligence System - Individual Reflection and Analysis

1. Project Organization and Management

I completed this project independently, which meant I was responsible for organizing the work, developing the code, testing the results, and troubleshooting problems. The project involved several Natural Language Processing (NLP) techniques, including text preprocessing, classification, sentiment analysis, named entity recognition, and system integration.

I used Google Colab to develop and test my notebook and GitHub to organize my work and track changes. I learned that keeping the code organized and testing each section before moving on to the next made a big difference. Working independently also helped me improve my time management, problem-solving skills, and confidence in handling technical challenges.

2. Technical Integration Challenges

The most challenging part of this project was getting the different NLP techniques to work together in one system. I worked with the BBC News dataset, which contained 1,490 training articles across five categories: Business, Entertainment, Politics, Sport, and Technology.

I explored TF-IDF, sentiment analysis, part-of-speech tagging, named entity recognition, and machine learning classification. One problem I encountered was that the classification model expected 5,006 features, while the TF-IDF vectorizer produced only 5,000. I had to troubleshoot the issue to make the model and vectorizer compatible.

3. Data Quality, Model Performance, and Lessons Learned

During my analysis of the BBC News dataset, I identified 50 duplicate articles among the 1,490 records. Although the dataset had no missing values, the duplicates highlighted the importance of checking data quality before training machine learning models. I retained the original dataset for this project, but in a future version I would remove duplicate articles before splitting the data into training and testing sets. This would help reduce the possibility of repeated articles affecting the evaluation results.

Another important lesson involved classification performance. My TF-IDF-only Logistic Regression model achieved approximately 96.98% accuracy in an initial evaluation. However, when I combined TF-IDF with additional features, the results were not as strong. The classifier comparison also showed that SVM achieved approximately 35.57% accuracy with the combined features.

These results taught me that adding more features does not automatically improve a machine learning model. How the features are prepared, scaled, and combined can significantly affect performance. I would like to improve this in the future by testing different feature combinations, applying feature scaling, and evaluating the models using a consistent training and testing process. Overall, this project helped me understand that developing an NLP system involves more than getting the code to run.

3. Business Value Assessment

I can see several practical business applications for a system like NewsBot. Organizations deal with large amounts of news and written information, and reviewing everything manually takes time.NewsBot could help automatically categorize articles, identify important people and organizations, and analyze sentiment.

It could also support business intelligence, competitive analysis, regulatory monitoring, and risk management.

With my background in the energy industry and risk management, I can see how a similar system could help companies monitor regulatory developments, competitors, and emerging business risks. I believe the biggest benefit would be identifying important information more quickly while still relying on human judgment for major decisions.

4. Individual Contributions

Since I completed the project independently, I was responsible for the different stages of the analysis and development.My work included exploring the dataset, preprocessing text, creating TF-IDF features, testing classification models, analyzing sentiment, and working with named entity recognition and part-of-speech tagging. I also created visualizations and worked on integrating the different components into the NewsBot system.

A significant amount of my time was spent troubleshooting the notebook. I worked through problems involving incorrect column names, missing variables, indentation errors, and model compatibility. This gave me valuable experience solving problems and helped me become more comfortable using Python and Google Colab.

5. Future Enhancements

There are several improvements I would like to explore in the future. I would be interested in testing newer NLP models and comparing their results with traditional machine learning approaches.

I would also like to improve how features are combined, use more recent news articles, and include publication dates so the system could identify trends over time.Another useful improvement would be creating a simple interface where someone could enter a news article and receive a category prediction, sentiment analysis, and a list of important entities. I think these changes could make NewsBot more useful in a real-world business setting.

6. Professional Development Impact

This project gave me valuable hands-on experience with Python, Google Colab, machine learning, and NLP libraries. I also gained a better understanding of how important data preparation, testing, and troubleshooting are when developing an AI application.One result that stood out to me was that the TF-IDF-only Logistic Regression model achieved approximately 97% accuracy in an initial evaluation, while adding additional features reduced accuracy to approximately 66%. This taught me that adding more features does not necessarily improve a model.

Coming from an engineering and business background, I enjoyed seeing how AI could be applied to real-world problems. The project helped me connect classroom concepts with practical business applications and increased my interest in continuing to learn about artificial intelligence. Overall, I feel more confident working with NLP techniques, troubleshooting code, and understanding how different components can be combined into a useful system.